

Question	Answer	Marks							
7(a)	charge / potential (difference)	M1							
	charge is charge on one plate, <u>and</u> potential is p.d. between the plates	A1							
7(b)(i)	straight line starting at the origin	B1							
	line with positive gradient ending at (V, Q)	B1							
7(b)(ii)	work done is the area under the graph	B1							
	$W = \frac{1}{2}QV$	A1							
7(c)(i)	final p.d. shown as $V/4$ for both capacitors	B1							
	final charges add together to give Q	B1							
	charge on Y = $3 \times$ charge on X (and both charges shown as a multiple of Q)	B1							
	Fully correct answer: <table border="1"> <tr> <td></td><td>X</td><td>Y</td></tr> <tr> <td>final p.d.</td><td>$V/4$</td><td>$V/4$</td></tr> <tr> <td>final charge</td><td>$Q/4$</td><td>$3Q/4$</td></tr> </table>			X	Y	final p.d.	$V/4$	$V/4$	final charge
	X	Y							
final p.d.	$V/4$	$V/4$							
final charge	$Q/4$	$3Q/4$							
7(c)(ii)	less than	B1							

9702/42

Cambridge International AS & A Level – Mark Scheme

October/November 2024

PUBLISHED

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